Nicolas Saikaly

Software Engineering Student · AI / Python

Paris area, France | +33 6 98 57 41 08 | saikalynicolas@gmail.com

github.com/NicolasSaikaly | linkedin.com/in/nicolas-saikaly

PROFILE

4th-year engineering student in Computer Science & Applied Mathematics, graduating 2028. Software engineer at **CEA**, France's leading public research organization, through a 3-year work-study program, delivering C++ and Python features for **SALOME**, an open-source simulation platform co-developed with EDF. Seeking an internship in AI / ML engineering to broaden my experience. **Available for 6–12 weeks between June and early September 2027.**

EXPERIENCE

CEA Paris-Saclay – French Alternative Energies and Atomic Energy Commission **Sep 2025 – Present**
Software Engineer, Work-Study (3 years, alternating between university and CEA) *Gif-sur-Yvette, France*

- **Made polyhedral meshing usable end-to-end in SALOME:** developed a robust C++ converter and a unified OpenFOAM export preserving all mesh groups, so engineers go straight from meshing to CFD simulation.
- **Improved mesh reliability:** diagnosed connectivity defects in simplified polyhedral meshes and fixed a gap in the library's repair step (covered by C++ unit tests), producing simulation-ready meshes.
- **Gave users direct control over mesh generation:** exposed new options of the Geogram and cfMesh meshers in the Qt interface (seeding strategy, multi-mesh selection, cell-size bounds) and made the plugin cross-platform.
- **Enabled automation and reproducibility:** exposed boolean mesh operations through a Python API and study dump so studies can be scripted and replayed; added cancellation and unit tests.
- **Strengthened the CFD chain:** fixed the loss of mesh groups in the visualization module and began integrating EDF's Code_Saturne boundary-layer insertion, preserving boundary conditions and near-wall accuracy.

NASCO Insurance Group – IT Department **Jun 2025 – Jul 2025**
IT Intern *Beirut, Lebanon*

- Learned **Angular** through internal training and built several web apps (TypeScript) in an English-speaking team.

EDUCATION

Polytech Paris-Saclay – Université Paris-Saclay | Work-study **Sep 2025 – Aug 2028 (expected)**
Engineering Degree, Computer Science & Applied Mathematics – currently in 4th year *Orsay, France*

- **Relevant coursework:** Machine Learning, Data Engineering, Algorithms & Data Structures, Software Testing.

ECAM-EPMI – IPSL **Sep 2023 – Jul 2025**
Intensive Preparatory Program in Mathematics & Physics – ranked 4th in class (year 1) *Cergy, France*

Lebanese Baccalaureate – General Sciences **Jul 2023**
Graduated with highest honors (16.5/20) *Lebanon*

SELECTED PROJECTS

SALOME Mesh Assistant – local RAG on technical docs (ongoing) | *Python* github.com/NicolasSaikaly/salome_assistant

- **Built a CPU-only RAG assistant** that answers SALOME meshing questions, in English or French, with cited documentation sources: section-aware chunking, embeddings + Chroma vector search, local LLMs via Ollama, Streamlit chat UI.
- **Let measurement drive the design:** built a retrieval benchmark (hit@k, MRR) to compare embedding models, rerankers and LLM query rewriting, reaching nearly 90% top-5 accuracy on CPU in a fraction of a second.

C-- Compiler | *C++*

- Writing from scratch a compiler for a subset of C producing executable output (lexing, parsing, code generation).

TECHNICAL SKILLS

Languages:	C++, Python, C, Java, SQL, TypeScript
AI / ML:	RAG, embeddings, vector search (Chroma), local LLMs (Ollama), retrieval evaluation, Streamlit
Engineering:	Unit testing (Python unittest, C++), debugging, API design, cross-platform development
Tools:	Linux, Git/GitHub, CMake, Qt, Angular, SALOME, ParaView

LANGUAGES & INTERESTS

Languages:	English (advanced, professional) · French (fluent) · Arabic (native)
Interests:	Football, basketball, padel, bodybuilding, travel, swimming